

Slide 1: Introduction to Machine Learning

Machine Learning (ML) is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed.

Key concepts covered in this course:

- Supervised Learning
- Unsupervised Learning
- Neural Networks
- Deep Learning

Slide 2: Supervised Learning

Supervised learning uses labeled datasets to train algorithms to classify data or predict outcomes.

Common algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines (SVM)
- Random Forest

Applications: spam detection, image classification, price prediction.

Slide 3: Unsupervised Learning

Unsupervised learning finds hidden patterns in data without labeled responses.

Types of unsupervised learning:

- Clustering (K-means, Hierarchical)
- Dimensionality Reduction (PCA, t-SNE)
- Association Rules

Applications: customer segmentation, anomaly detection, recommendation systems.

Slide 4: Neural Networks

Neural networks are computing systems inspired by biological neural networks in the brain.

Architecture components:

- Input Layer: receives initial data
- Hidden Layers: process and transform data
- Output Layer: produces final predictions
- Weights and Biases: adjustable parameters
- Activation Functions: add non-linearity

Slide 5: Summary and Next Steps

In this module, we covered the fundamentals of machine learning:

- Understanding ML types: supervised vs unsupervised
- Key algorithms and their applications
- Introduction to neural network architecture

Next module: Deep Learning and Convolutional Neural Networks (CNNs).

Questions? Contact: instructor@university.edu